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CURRENT OPINION

The Dose–Response Relationship of Exercise-Based Injury Prevention Programmes: Implications for Research and Practice

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Abstract

Exercise-based injury prevention programmes (EIPPs) need a certain period of consistent practice to achieve the expected efficacy. Therefore, as with many other biomedical interventions, EIPPs can be characterised by a dose–response relationship. In this Current Opinion, we aim to present the concept of the dose–response relationship and explore the research and clinical implications stemming from it. First, we introduce its origins from pharmacology and define the terms dose and response in the context of EIPPs. Specifically, we describe how the dose is dependent on the duration of EIPP practice and how the response can be quantified from injury-related epidemiological outcomes. Then, we highlight that the vast majority of research studies examining the efficacy of EIPPs neglect their dose–response relationship, which can lead to a potential underestimation of the EIPP effect. To overcome this problem, we present two statistical approaches that were applied to a previous study, providing a practical way to estimate the dose–response relationship of an EIPP. Finally, we discuss how researchers can integrate this concept in their study designs and analyses and how practitioners can plan the implementation of EIPPs on the basis of the time needed for the EIPPs to gain efficacy according to the dose–response relationship.

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Key Points

Exercise-based injury prevention programmes need a sufficient period of time (i.e. duration in days or weeks) to elicit structural and/or functional adaptations and ultimately to gain efficacy for injury-risk reduction. This can be described by a dose–response relationship, where dose refers to the duration of the period that the exercise-based injury prevention programme is practised.

Research studies exploring the efficacy of exercise-based injury prevention programmes to reduce injury risk often neglect the dose–response relationship by analysing injuries sustained over the entire follow-up from the start of the intervention period. This can lead to an underestimation of the effect of the programme.

There are statistical ways to integrate the dose–response relationship in the analysis of efficacy studies of exercise-based injury prevention programmes which can avoid the underestimation of their effect. The same methods can help estimate and visualise the dose–response relationship of a given programme.

Understanding the dose–response relationship of exercise-based injury prevention programmes can help researchers to avoid underestimating their effect during analysis of randomised controlled trials and practitioners to better plan their implementation according to the targeted period when maximal efficacy may be needed.

1 Introduction

Exercise-based injury prevention programmes (EIPPs) have been extensively studied and applied as an important injury-risk reduction strategy in sports [1–4]. However, an often-neglected aspect of EIPPs is understanding how the EIPP effect on injury risk gradually increases as practice time progresses. This concept can be defined as the dose–response relationship of EIPPs. In this Current Opinion, we aim to discuss the research and clinical implications stemming from the concept of the dose–response relationship of EIPPs. For that, we first present conceptual considerations of the dose–response relationship and frame it in the context of EIPPs. Then, we highlight the potential risks of not acknowledging this relationship in research, we show that current research practices do not account for these risks and we propose statistical solutions for successful integration of the dose–response relationship in randomised controlled trials (RCTs). Finally, we discuss the research and clinical/practical implications emerging from integrating this concept.

2 What is a Dose–Response Relationship?

The dose–response relationship was originally conceived in physiology and is of fundamental importance in pharmacology and toxicology [5]. It aims to associate any amount of an intervention with a certain level of change in an outcome (i.e. effect) [5]. Dose–response relationships are fundamental for the development of safe and effective drugs, as they can inform on their therapeutic and toxic dose ranges [6]. Consequently, investigating the dose–response of a new intervention is one of the objectives of early phase clinical trials (phase I–IIa), and it precedes the execution of efficacy and effectiveness trials (phase IIb–III) [7]. Dose–response relationships can be typically modelled using the 'Hill equation' (Equation 1), first proposed by A.V. Hill when he described the association between oxygen pressure and haemoglobin saturation, and mathematically expressed as [8]:

$$\frac{\text{Response}}{\text{Response}_{\text{max}}} = \frac{\text{Dose}^{a}}{\text{ED}_{50}^{a} + \text{Dose}^{a}} \quad (1)$$

where $\text{Response}_{\text{max}}$ is the largest possible response that can be achieved (either expected or measured), ED_{50} is a constant representing the value of dose for which 50% of the maximum response can be achieved and a is the Hill coefficient representing the synergistic ($a > 1$) or antagonistic ($a < 1$) interaction of increasing quantities of dose and which characterises the 'sigmoidicity' of the curve (Fig. 1A) [8, 9].

Equation 1 can also describe the pharmacodynamics (i.e. what a drug does to the body) [10] of any treatment or toxic chemical substance [5, 8]. This concept can be extended to other forms of interventions that interact with the human body and elicit a biological response, such as physical activity or exercise [11, 12], nutritional interventions [13], psychiatric interventions [14, 15] and toxicity by ionising radiation [16].

Regardless of the type of intervention, a challenge of describing the dose–response relationship lies in the definitions of dose and response. Typically, dose is the amount of the agent that interacts with the organism [17]. However, the definition is context-dependent. For example, the dose of a medical drug could be defined either as the prescribed and ingested dose (i.e. external dose) or as the concentration of the drug in the blood or the target tissue (i.e. internal dose). Similar distinctions have been applied in sport science to define training load [18]. In the context of EIPPs, the most convenient definition of dose is that of external dose, which is the product of the dose of an individual session multiplied by the total number of

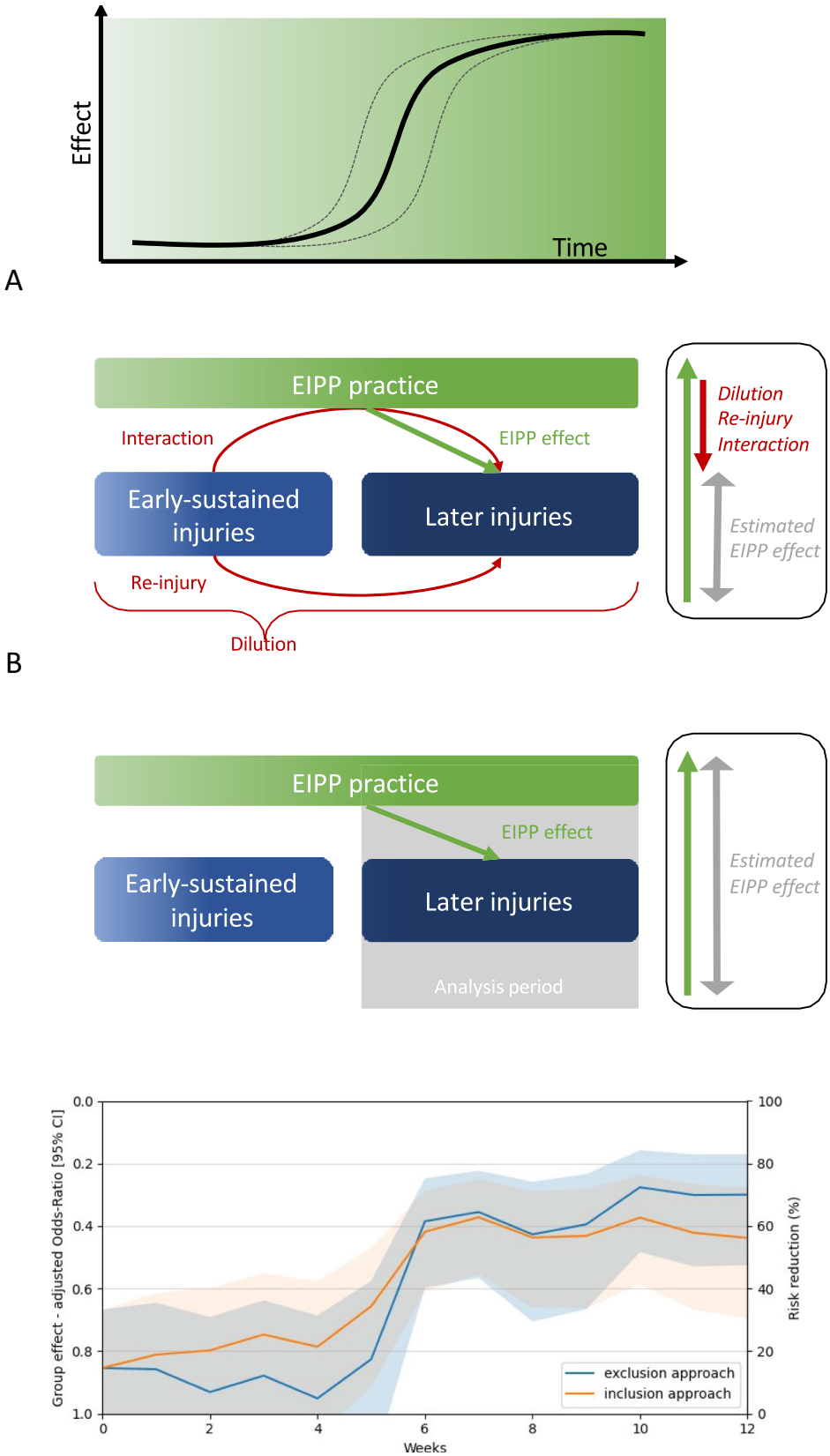

Fig. 1 (**A**) The dose–response relationship of an exercise-based injury prevention programme (EIPP). (**B**) Including in the analysis the period before the EIPP achieves efficacy introduces negative effects (dilution, re-injury and interaction effects) which reduce the estimated effect. (**C**) Focusing the analysis on the period after the EIPP achieved efficacy while adjusting for the re-injury and interaction effects can better isolate the real effect of the EIPP (adapted from Iatropoulos et al. [27]). (**D**) The dose–response relationship of an EIPP in athletics estimated with the exclusion and the inclusion statistical approaches with notable agreement between the two methods. The odds ratio between the intervention and control group with respect to the weekly prevalence of injuries is plotted with respect to the length of the period excluded from the analysis. For more information, see Iatropoulos et al. [27]

sessions performed (Dose = $\text{Dose}_{\text{session}} \times N_{\text{sessions}}$). The dose of an individual session can be quantified as the product of a measure of volume, such as the number of exercises performed or the time needed to be completed, and a measure of intensity, such as the rate of perceived exertion ($\text{Dose}_{\text{session}}$ = Volume × Intensity) [4, 5]. Similarly, the total number of sessions performed can be quantified as the product of the frequency of sessions performed every week and the total duration of the period of EIPP practice (N_{sessions} = Frequency × Duration) [4, 5]. Thus, the definition of dose of an EIPP can be mathematically expressed as:

$$\text{Dose} = \text{Volume} \times \text{Intensity} \times \text{Frequency} \times \text{Duration} \quad (2)$$

Based on Eq. 2, a dose–response relationship can be challenging to describe when each of the four parameters of dose can be modified simultaneously. In a simpler scenario in which the parameters are independent (i.e. no synergistic or impeding interaction between them), if any three parameters remained fixed, it would be easier to describe the dose–response relationship by assessing the influence of the fourth parameter on the response. To a certain degree, this could apply to EIPPs, which often have a fixed volume, intensity and frequency in research and/or field settings (e.g. Fédération Internationale de Football Association [FIFA] 11+ programme) [3]. Thus, as duration increases, dose increases proportionally, which renders the dose–response relationship equivalent to a duration–response relationship.

Respectively, the response can be defined as a biological adaptation, an epidemiological outcome of a disease or an appropriate surrogate metric [5, 18]. In the context of EIPPs, biological adaptations could be quantified by the absolute or relative change of a parameter, such as an anthropometric characteristic, a biomechanical property, a structural measurement from imaging techniques or a biochemical substance. Such outcomes are preferred when we want to explain the mechanism behind the effect of injury-risk reduction of EIPPs. Instead, if we seek to explore how the effect of EIPPs on injury-risk reduction is dependent on the dose, injury-related epidemiological outcomes such as the prevalence, incidence or burden should be preferred. For example, in RCTs evaluating the efficacy of EIPPs to reduce injury risk, the response can be estimated as the relative or absolute difference in an injury-related epidemiological outcome between the intervention group practicing the EIPP and the control group. Lastly, surrogate outcomes could be biological parameters or a combination of parameters (e.g. clinical score) which are linked to the actual injury risk or severity. Surrogate outcomes can be used to quantify response when there is no information on the actual outcome of interest (i.e. epidemiological outcome) [19].

3 Is There a Dose–Response Relationship of EIPPs?

Intuitively, EIPPs need to be performed for a sufficient period of time (i.e. duration in days or weeks) before they elicit their protective effect to reduce sports injuries, a 'doctrine' stemming from the seminal work of Selye on the 'general adaptation syndrome' of physiology [20]. Mechanistically, EIPPs can exert their preventive effect through structural and functional adaptations [21]. Such adaptations have been shown to demand a certain period of consistent practice to emerge [22–24]. For example, eccentric training has led to improved strength, increased fascicle length, decreased pennation angle and serial addition of sarcomeres in the hamstring muscles within 9–36 weeks [22, 23]. In addition, the stiffness and echointensity of the Achilles tendon improved gradually over 3 months of isometric strength training of the plantar flexor muscles [24]. Depending on the anatomical location and mechanism of injury that an EIPP is intended to prevent, different adaptations can be measured. Such adaptations may elucidate the biological basis of the dose–response relationship of the examined EIPP, but they remain surrogate outcomes to the more relevant outcome of injury-risk reduction. Only by using injury-related epidemiological outcomes can we describe the dose–response relationship of EIPPs in a way that the response reflects the effect to reduce injury risk. For example, a study compared two different volumes of the same EIPP, while controlling for frequency and duration, on the reduction of injury rates in youth footballers [25].

To our knowledge, no study has examined the dose–response relationship of EIPPs when the dose is defined as the duration of EIPP practice (i.e. controlled for volume, intensity and frequency) and the response as an injury-related epidemiological outcome. Only one meta-analysis of RCTs testing the efficacy of different EIPPs analysed the associations between the dosing scheme of EIPPs and injury-risk reduction [4]. Specifically, Steib et al. [4] categorised the volume, frequency and duration of several EIPPs in two to three categories each and compared the injury-risk reduction between categories. The authors concluded that higher volumes and frequencies were associated with larger risk reduction effects [4]. However, by categorising the EIPPs by short (1–6 months) and long (7–12 months) duration, it was not possible to draw precise conclusions on how the EIPP effect may change as duration increases. The study simply observed no difference in injury-risk reduction between shorter and longer durations [4].

This lack of direct evidence is not surprising. A straightforward study design to determine the dose–response

relationship would be to assign a different duration of EIPP practice to different groups and then compare the injury rates between groups over a follow-up period long enough to allow for injuries to accumulate. However, the sample size requirements for such studies could be discouraging. Alternatively, as sometimes proposed in pharmacological studies, the same group could perform the EIPP for different durations with the same follow-up while ensuring adequate wash-out periods before starting another period of EIPP practice. Here, the disadvantages would be an extremely long study period and a potential interaction of the sequential rounds of EIPP practice and/or the injuries sustained during each round. Despite these challenges in estimating the dose–response relationship, we should be aware of its existence during the design and analysis of research studies exploring the efficacy of EIPPs.

4 Are We Underestimating the EIPP Effect?

Because of the lack of direct evidence of the dose–response relationship of EIPPs, most RCTs apply the EIPP for very long periods (longer than 10 weeks), which increases the probabilities that EIPPs will gain effect at some point. However, the period of injury data collection and analysis of injury risk starts at the same time as the intervention is applied and usually ends at the same time as the EIPP [1, 26]. This approach is mathematically equivalent to assuming that the efficacy of the EIPP to reduce injury risk is achieved as soon as it starts, and ignores its dose–response relationship, which, as shown earlier, is based on the time needed for the structural and functional adaptations to emerge [21–24]. Indeed, any injury sustained before the EIPP becomes efficacious (i.e. early sustained injury) could not have been possibly prevented by the EIPP. Consequently, by statistically analysing early sustained injuries in the same way as injuries sustained after the EIPP has gained its potential efficacy, we may introduce bias in the estimation of this efficacy [27]. Of note, this bias can be proportional to the relative size of the period needed for the EIPP to gain efficacy compared with the total follow-up period. Thus, since most studies apply EIPPs for long periods, the effect could still be estimated as significant [1–3, 28]. However, this cannot exclude the possibility that the estimated effect could have been even larger had the dose–response relationship been considered. Moreover, some of the studies which did not find a significant risk reduction effect could have missed it by not considering this dose–response relationship [27].

More concretely, when the analysis period starts at the same time as the intervention starts, three interfering effects can lead to an underestimation of the EIPP effect (Fig. 1B) [27]. First, given that before the EIPP becomes efficacious, the two study groups in an RCT (intervention and control) have the same risk of injury, early sustained injuries would increase the total number of injuries in both groups and hence dilute any potential difference between them (dilution effect) [27]. As an example, a study evaluating a 13-week Nordic hamstring exercise programme in football players showed a 67% reduction in hamstring injury prevalence after the end of the intervention period (weeks 13–52), but over the entire follow-up period (weeks 1–52), including the injuries sustained during the 13-week intervention period, the prevalence reduction would be diluted to 56% (Table 1) [29]. Second, since previous injuries have been identified as a risk factor for subsequent injuries, athletes with an early sustained injury may be at a higher risk of injury for the remaining follow-up period (re-injury effect) [27]. Although it could be argued that this re-injury risk would be the same between the injured athletes of both groups, it is through a similar dilution process with additional injuries being sustained from groups that the re-injury effect may decrease the observed effect of the EIPP. Third, it is possible that an early sustained injury may attenuate the subsequent adaptations of the EIPP (interaction effect), either because this injury interferes with the elicited adaptations of the EIPP or because of behavioural reasons (e.g. injured athletes may pause performing the EIPP until they return to training) [27]. This effect only affects athletes in the intervention group. Collectively, these three effects may underestimate the EIPP effect, even to the point of claiming that an EIPP is not efficacious when it actually is [27].

5 Current Research Practices

The potential underestimation of the EIPP effect could be one of the reasons why there is mixed evidence regarding the efficacy of EIPPs. Specifically, there are three recent RCTs [30–32] and one meta-analysis [26] concluding that the proposed EIPPs had no effect in reducing injury risk. Meanwhile, meta-analyses that have concluded that EIPPs had a risk reduction effect included several individual studies which did not report this effect [1–4]. Although this lack

Table 1 Hamstring muscle injuries sustained per group in each period of the study by Van der Horst et al. [29]

	Intervention group ($n = 292$)	Control group ($n = 287$)	Relative prevalence
Intervention period (weeks 1–13)	5 (1.7%)	7 (2.4%)	0.71
Analysis period (weeks 14–52)	6 (2.1%)	18 (6.3%)	0.33
Total period (weeks 1–52)	11 (3.8%)	25 (8.7%)	0.44

of efficacy could be attributed to the characteristics of the EIPPs, it is also possible that the EIPPs were actually efficacious, but the design and analysis of the studies could not have been entirely adequate to capture this. Specifically, current research practices may not consider the dose–response relationship of EIPPs during the statistical analysis and the related interfering effects (i.e. dilution, re-injury and interaction). For example, the meta-analysis by Wu et al. [26] included only studies that started the analysis period simultaneously with the intervention period, thus not acknowledging the dose–response relationship of EIPPs.

To understand the degree of integration of the dose–response relationship concept in the field of sports injury prevention, we performed a non-systematic screening of literature and collected all the included studies of six recent systematic reviews analysing the efficacy of EIPPs on injury-risk reduction [1, 2, 4, 28, 33, 34]. Specifically, for each study, we checked whether and to what extent the design and analysis accounted for the dilution, re-injury and interaction effects previously described. After removing duplicates and retaining only RCTs in sport populations, 36 studies were screened. Only one study intentionally separated the intervention period from the analysis period during statistical analysis [29], and two other studies performed an unintentional separation owing to the study design [35, 36]. Thus, only these three studies accounted for the dilution effect but not for the re-injury and interaction effects. The remaining 33 studies (92%) did not account for any of the three effects in any way. To our knowledge, only one more recent RCT, not included in the abovementioned meta-analyses, intentionally separated the intervention and the analysis period, while indirectly pointing to the dose–response relationship in their justification and found a large effect of that EIPP [37].

The fact that the vast majority of RCTs evaluating EIPPs did not acknowledge the dose–response relationship of EIPPs in their analysis can have an impact on the validity of their results and, therefore, on the appropriateness of the evidence-based recommendations regarding the practice of EIPPs. Therefore, it is worth discussing potential statistical solutions to adequately consider the dose–response relationship of EIPPs in the analysis of RCTs.

6 Statistical Integration of the Dose–Response Relationship

To control for the three interfering effects (dilution, re-injury and interaction), we suggest two distinct approaches for the statistical analysis that we define as the (i) inclusion approach, in which all athletes are included in the analysis, and (ii) exclusion approach, in which athletes injured before the start of the analysis period are not included (Table 2) [27].

In both approaches, we first address the dilution effect by including in the analysis only injuries sustained after the time of expected efficacy (Fig. 1C). To our knowledge, only two RCTs evaluating EIPPs have included such considerations in their primary analyses [29, 37]. In our secondary analysis of an RCT, which reported no EIPP efficacy in the primary analysis including injuries over the whole intervention period [30], we found that suppressing the dilution effect by including in the analysis only the injuries sustained after the first few weeks improved the effect of the EIPP, with the largest improvement observed at 7 weeks {from an odds ratio with 95% confidence interval (OR [95%CI]) of 0.85 [0.67-1.09] to 0.75 [0.56–1.01]} [27]. Further controlling for the re-injury and interaction effects improved the effect of the EIPP even more. Specifically, in the inclusion approach, we controlled for the re-injury effect by adjusting the statistical analysis for the occurrence of an early sustained injury and for the interaction effect by adjusting for the interaction term between the allocation group (intervention/control) and the occurrence of an early sustained

Table 2 Comparison of the described statistical approaches integrating the dose–response relationship of EIPPs

Statistical approach	Controlled effects	Initial period	Included athletes	Modelling formula and adjusting variables	EIPP effect (OR [95% CI])
Primary analysis	None	Included	All	$Y \sim$ Gr + Conf	0.85 [0.67–1.09]
First step	Dilution	Excluded	All	$Y \sim$ Gr + Conf	0.75 [0.56–1.01]
Exclusion approach	Interaction, re-injury and dilution	Excluded	Injury-free during the initial period	$Y \sim$ Gr + Conf	0.28 [0.16–0.48]
Inclusion approach	Interaction, re-injury and dilution	Excluded	All	$Y \sim$ Gr + Conf + EI + (Gr × EI)	0.37 [0.25–0.55]

For each approach, the effects which each model controls for, the included periods and the included athletes are presented, as well as the model's variables and the EIPP effect estimated with the approach in the secondary analysis of Iatropoulos et al. [27]

OR odds ratio, *Y* weekly ICPR (outcome), *Gr* group variable (intervention/control), *Conf* the common confounders as per the original study (past season ICPR + Compliance + Exposure), *EI* early sustained injury (binary), *(Gr × EI)* the interaction term between the group and early sustained injuries (adapted from Iatropoulos et al. [27])

injury (Table 2). Hereby, we observed a larger EIPP effect (OR [95%CI] = 0.37 [0.25–0.55]) [27]. Alternatively, with the exclusion approach, which also controlled for the three effects (Table 2), we observed similar improvements of the EIPP effect (OR [95%CI] = 0.28 [0.16–0.48]) [27]. Researchers in other fields which test preventive interventions such as vaccines have already adopted considerations similar to the exclusion approach in their analyses [38]. To our knowledge, no primary analysis of an RCT exploring the efficacy of an EIPP has ever addressed the re-injury and interaction effects.

As detailed by Iatropoulos et al. [27], if the optimal period between the start of the intervention and the start of the analysis is unknown, we can repeat the analysis using either approach for a range of possible period lengths. Then, by plotting the estimated intervention effect for the different lengths of the excluded period, we can visualise the dose–response relationship of the EIPP (Fig. 1D) [27]. This could be a practical way to describe this relationship straightforwardly, avoiding very long follow-up periods or large sample size requirements.

The agreement of the results between the inclusion and the exclusion statistical approaches in our study [27] is a confirming indicator of the theoretical basis of the dose–response relationship of EIPPs (Fig. 1D). Of course, replication of the analyses with other datasets of RCTs studying EIPPs is needed to validate the approach. As a future perspective, Eq. 1 could be used to model the dose–response relationship of an EIPP to determine the values of ED_{50} and a, which can facilitate comparisons between different studies.

7 Implications for Research and Clinical Practice

There are several research and clinical implications stemming from the concept of the dose–response relationship of EIPPs (Table 3).

Researchers can have the opportunity to estimate the dose–response relationship of an EIPP prior to exploring its efficacy. Alternatively, if they perform an RCT to investigate the efficacy of an EIPP, they can also estimate its dose–response relationship as a supplementary analysis. In such studies, we recommend not including in the analyses those injuries sustained during the initial intervention period when the EIPP has not yet gained full efficacy. As mentioned above, the length of this initial period can be informed by the previous estimation of the dose–response relationship of the EIPP or by experience. Researchers can also consider whether the injuries sustained during the initial intervention period can mitigate the efficacy of the EIPP and, if so, they can adjust

Table 3 Research and clinical implications of the dose–response relationship of EIPPs for different stakeholders of injury prevention

	What should they do	What should they avoid
Researchers	- If possible, estimate the dose–response relationship of an EIPP prior to exploring its efficacy - Alternatively, consider estimating the dose–response relationship of an EIPP as a supplementary analysis of the study exploring its efficacy, and include this in the study's protocol - Injuries sustained during the initial intervention period should be excluded from the analysis as they occurred before the EIPP gained efficacy. The length of this initial period can be informed by the estimation of the dose–response relationship or experience - Consider whether these injuries sustained during the initial intervention period can mitigate the efficacy of the EIPP, and adjust this analysis accordingly	- Analysing injury outcomes from the start of the intervention period
Practitioners and Coaches	- Identify how much time is needed for an EIPP to gain efficacy from the dose–response relationship. Use this information to determine how early to start the EIPP to achieve maximal efficacy for the targeted period when maximal efficacy is needed - When reading the available evidence, be cautious about the results of studies that analyse injury outcomes from the start of the intervention period	- Expecting an injury-risk reduction as soon as they implement an EIPP
Athletes	- Practise the EIPP consistently for a period of time to gain the risk reduction effect. Plan this period earlier than the targeted period when maximal efficacy is needed on the basis of the dose–response relationship - Be especially cautious during the first weeks of practising an EIPP, as an injury could mitigate the effect of the EIPP later	- Expecting an injury-risk reduction as soon as they implement an EIPP - Being discouraged to continue practising the EIPP if they sustain an injury soon after the start of the EIPP practice. This injury may have occurred before the EIPP gained efficacy

EIPP exercise-based injury prevention programme

the analysis accordingly. The results of previous RCTs can be re-analysed with the proposed approaches to estimate the dose–response relationship of the EIPP retrospectively. Such prior knowledge can inform the study design of future studies.

Coaches and sport practitioners should not expect that EIPPs have an immediate injury-risk reduction effect. Knowing the dose–response relationship of an EIPP could help them optimise the planning of its implementation. Specifically, they could determine more accurately when to start the EIPP to reach efficacy for the targeted period in which maximal efficacy is needed (e.g. competition period). In our example of an EIPP in athletics, 6 weeks were necessary before the programme gained full efficacy [27]. However, this period could vary among EIPPs and sports. Moreover, when reading the available evidence, caution is needed in interpreting the results of studies that analyse injury outcomes from the start of the intervention period.

Finally, athletes should be aware that EIPPs need a period of consistent practice to achieve a potential risk reduction effect. In addition, if athletes sustain an injury soon after the start of the EIPP, they should not be discouraged from continuing its practice, since this injury may have occurred before the EIPP gained efficacy. However, athletes should be aware that such early injuries are especially undesirable as they can mitigate the effect of the EIPP later.

8 Perspectives

This discussion stressed the importance of considering the duration of the period of EIPP practice needed to gain efficacy for the design and analysis of research studies and for clinical practice on the field. However, a holistic approach to the dose–response relationship of an EIPP would need this relationship to be described for each of the four fundamental components of dose (i.e. volume, intensity, frequency and duration) separately and jointly, which would warrant highly sophisticated study designs [4]. Future research should address aspects related to the precise estimation of dose in a sports injury prevention context, such as (i) the source of data collection (e.g. reported by athletes or supervising personnel), (ii) the calculation of volume per exercise (e.g. repetitions or time spent) and EIPP volume (e.g. total time per session or number of exercises), (iii) the calculation of total dose when volume, intensity and/or frequency change over the entire period and (iv) the influence of supervision, adherence and fidelity to the EIPP [39, 40].

9 Conclusions

The dose–response relationship is a concept that can be extended from pharmacological treatments to interventions such as EIPPs. Currently, in research, this relationship is not usually considered in the analysis of RCTs that explore the efficacy of EIPPs, which can often lead to an underestimation of their effect. We presented two statistical approaches to estimate and visualise this relationship, which can be used during the design and analysis of studies exploring the efficacy of EIPPs on injury-risk reduction. Researchers are advised to avoid analysing injury outcomes from the start of the intervention period of an EIPP. Coaches, practitioners and athletes should plan the consistent practice of EIPPs several weeks before the period in which they aim to lower injury risk (e.g. competition). All sports stakeholders should avoid determining the effect of EIPPs on the basis of the injuries occurring early after initiating the practice of these programmes. EIPPs need time to 'honour their name'.

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Declarations

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